

300+ SAP ABAP

INTERVIEW Q&A

THE ONLY SAP ABAP INTERVIEW BOOK YOU'LL EVER NEED.

Topic wise Question

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By Bhagwat Bhokare

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1. Basic questions

1. What is SAP ABAP?

Answer:

ABAP stands for *Advanced Business Application Programming*. It is SAP's primary programming language used to develop business applications in the ABAP runtime environment. It is tightly integrated with SAP modules like MM, SD, FI, PP, and works directly with SAP tables, Data Dictionary objects, and business processes.

ABAP is mainly used for custom reports, enhancements, interfaces, forms, workflows, and CDS-based development in modern S/4HANA systems. One of the key strengths of ABAP is that it is optimized for business logic and database handling inside the SAP system, which makes it different from general-purpose languages.

13. What is the difference between Parameters and Select-Options?

Answer:

Both PARAMETERS and SELECT-OPTIONS are used to create input fields on the selection screen, but they have different capabilities:

PARAMETERS

- Single-value input field.
- Can accept only one value at a time.
- Syntax example:

PARAMETERS: p_bukrs TYPE bukrs. "Company Code

- Best used for mandatory single inputs.

SELECT-OPTIONS

- Creates a range table automatically.
- Can accept single values, multiple values, ranges, and exclusions.
- Syntax example:

SELECT-OPTIONS: s_bukrs FOR bukrs.

- Best used when multiple selection criteria are required.

Key Difference Table:

Feature	PARAMETERS	SELECT-OPTIONS
Number of Values	Single	Multiple / Range
Data Type	Elementary only	Can generate internal table
Auto Range Table	No	Yes
Flexibility	Less	More

Example Use Case:

- PARAMETERS → Input one company code for a report.
- SELECT-OPTIONS → Input multiple company codes or a range for reporting.

2. Data dictionary

21. What is the ABAP Data Dictionary?

Answer:

The ABAP Data Dictionary (DDIC) is a central repository in SAP where all metadata of database objects are defined and managed.

It allows developers to define tables, views, data elements, domains, types, and structures without directly creating database tables.

Key Points:

- Ensures data consistency across SAP applications.
- Automatically generates database tables and structures in the underlying DB.
- Supports validation rules, search helps, foreign keys, and table buffering.
- Changes in DDIC are transported across systems via transports.

Example:

- KNA1 (Customer Master) and MARA (Material Master) are defined in DDIC.
- Data elements like KUNNR define the semantic meaning of a field (customer number) while the domain defines its technical attributes like length and type.

Key Point:

DDIC acts as a bridge between ABAP programs and the database, providing a consistent and centralized data definition.

4. String operations

61. What is CONCATENATE?

Answer:

In ABAP, CONCATENATE is used to combine two or more strings or fields into a single string.

Key Points:

- Can join variables, literals, or constants.
- Optional SEPARATED BY keyword to insert separators.
- Result is stored in a target variable.

Example:

```
DATA: lv_fname TYPE string VALUE 'Bhagwat',  
      lv_lname TYPE string VALUE 'Bhokare',  
      lv_fullname TYPE string.
```

```
CONCATENATE lv_fname lv_lname INTO lv_fullname SEPARATED BY space.  
WRITE lv_fullname. "Output: Bhagwat Bhokare"
```

CONCATENATE is simple and widely used for combining strings in reports, logs, or messages.

5. Internal tables, work areas & operations

76. What is an Internal Table?

Answer:

In ABAP, an Internal Table is a temporary, in-memory table used to store and process a collection of data during program execution.

Key Points:

- Exists only during program runtime.
- Useful for storing database query results, calculations, or processing large datasets.
- Types of internal tables: Standard, Sorted, Hashed.
- Always works with a work area for row-level operations.

Example:

```
TYPES: BEGIN OF ty_customer,  
       kunnr TYPE kunnr,  
       name  TYPE name1,  
END OF ty_customer.
```

```
DATA it_customers TYPE STANDARD TABLE OF ty_customer.  
DATA wa_customer TYPE ty_customer.
```

"Adding a record

```
wa_customer-kunnr = '1000'.  
wa_customer-name  = 'Bhagwat'.  
APPEND wa_customer TO it_customers.
```

Internal Tables are core to ABAP programming for data manipulation without touching the database repeatedly.

78. What are the types of internal tables?

Answer:

In ABAP, internal tables are classified into three types, each with specific characteristics and use cases.

1. Standard Table

- Default type of internal table.
- Unsorted, allows duplicate entries.
- Access by index.
- Commonly used when order is not critical.

2. Sorted Table

- Automatically sorted by a specified key.
- Allows duplicate entries unless UNIQUE key is specified.
- Faster read operations using binary search.
- Ideal when data needs to be sorted or searched frequently.

3. Hashed Table

- Access by primary key only.
- No index, lookup is very fast.
- Unique entries only.
- Suitable for large datasets where key-based access is needed.

Example Declaration:

```
TYPES: BEGIN OF ty_customer,  
       kunnr TYPE kunnr,  
       name  TYPE name1,  
END OF ty_customer.
```

```
DATA it_std  TYPE STANDARD TABLE OF ty_customer WITH DEFAULT KEY.
```

```
DATA it_sorted TYPE SORTED TABLE OF ty_customer WITH UNIQUE KEY kunnr.
```

```
DATA it_hash  TYPE HASHED TABLE OF ty_customer WITH UNIQUE KEY kunnr.
```

Key Point:

Choose table type based on access pattern, key uniqueness, and performance requirements.

106. Difference between PARAMETERS and SELECT-OPTIONS

Answer:

In ABAP, PARAMETERS and SELECT-OPTIONS are both used on selection screens to get input from users, but they have different purposes and features.

Comparison Table:

Feature	PARAMETERS	SELECT-OPTIONS
Purpose	Single value input	Single or range of values
Number of Values	Only one value	Can enter multiple values or ranges
Usage	Simple filters, single input	Complex filters, ranges, multiple selections
System Table Binding	Can be bound to a data element	Must be bound to a data element
Screen Layout	Single field	Two fields: Low and High, with multiple entries possible
Example Syntax	PARAMETERS p_cust TYPE kunnr.	SELECT-OPTIONS s_cust FOR kunnr.

Example:

```
REPORT z_sel_demo.
```

```
PARAMETERS p_cust TYPE kunnr. "Single customer
```

```
SELECT-OPTIONS s_date FOR sy-datum. "Date range selection
```

```
START-OF-SELECTION.
```

```
WRITE: / 'Customer:', p_cust.
```

```
LOOP AT s_date.
```

```
WRITE: / 'From:', s_date-low, 'To:', s_date-high.
```

```
ENDLOOP.
```

- PARAMETERS → for single input.
- SELECT-OPTIONS → for ranges, multiple entries, or dynamic selection, providing more flexibility on selection screens."

144. How to Upload Files Using GUI_UPLOAD?

Answer:

In SAP ABAP, GUI_UPLOAD is a standard function module used to upload files from the local frontend (user's PC) to an internal table in the ABAP program.

It is commonly used for reading CSV, TXT, or other text files for further processing.

Key Points:

- GUI_UPLOAD reads a file line by line into an internal table.
- Runs on the frontend (SAP GUI), not the application server.
- Syntax (classic usage):

CALL FUNCTION 'GUI_UPLOAD'

EXPORTING

filename = lv_file

filetype = 'ASC'

IMPORTING

filelength = lv_length

TABLES

data_tab = it_data

EXCEPTIONS

file_open_error = 1

file_read_error = 2

no_batch = 3

gui_refuse_filetransfer = 4

invalid_type = 5

no_authority = 6

unknown_error = 7

bad_data_format = 8

header_not_allowed = 9

separator_not_allowed = 10

filesize_not_allowed = 11

header_too_long = 12

dp_error_create = 13

dp_error_send = 14

dp_error_write = 15

unknown_dp_error = 16

access_denied = 17

dp_out_of_memory = 18

disk_full = 19

dp_timeout = 20

file_not_found = 21

```
datapvider_exception = 22
control_flush_error  = 23
OTHERS               = 24.
```

- Always check SY-SUBRC to handle errors.

Real-Time Example (txt Upload):

Upload a txt of customer details:

```
DATA: it_file TYPE TABLE OF string,
      lv_file TYPE string.
```

```
lv_file = 'C:\Users\PC\Desktop\customer.txt'.
```

```
CALL FUNCTION 'GUI_UPLOAD'
```

```
EXPORTING
```

```
  filename = lv_file
```

```
  filetype = 'ASC'
```

```
TABLES
```

```
  data_tab = it_file
```

```
EXCEPTIONS
```

```
  OTHERS = 1.
```

```
IF sy-subrc = 0.
```

```
  LOOP AT it_file INTO DATA(lv_line).
```

```
    WRITE: / lv_line.
```

```
  ENDLOOP.
```

```
ELSE.
```

```
  WRITE: / 'Error uploading file'.
```

```
ENDIF.
```

- The uploaded data can then be processed into structures or internal tables for further logic.

GUI_UPLOAD is ideal for frontend file upload scenarios, especially for small-to-medium text or CSV files in ABAP programs.

9. Reports

150. What is Classical Report?

Answer:

In SAP ABAP, a Classical Report is a type of ABAP program that is used to display data in a simple list format on the SAP GUI screen.

It is one of the most basic reporting methods in SAP and is widely used for listing, printing, or summarizing data.

Key Points:

- Data Retrieval: Usually fetches data from database tables using SELECT statements.
- Output: Displayed using WRITE statements on the standard list screen.
- Events: Controlled by events like INITIALIZATION, START-OF-SELECTION, END-OF-SELECTION, and TOP-OF-PAGE.
- No Advanced Features: Does not include ALV features like sorting, filtering, or interactive events by default.
- Simple Structure: Best for straightforward, read-only reporting.

Real-Time Example:

Classical report to display customer names and cities:

```
REPORT zcustomer_list.
```

```
DATA: it_customer TYPE TABLE OF zcustomer,  
      wa_customer TYPE zcustomer.
```

```
SELECT kunnr name1 city FROM zcustomer INTO TABLE it_customer.
```

```
WRITE: / 'Customer List'.
```

```
WRITE: / 'KUNNR', 10 'NAME1', 30 'CITY'.
```

```
LOOP AT it_customer INTO wa_customer.
```

```
  WRITE: / wa_customer-kunnr, 10 wa_customer-name1, 30 wa_customer-city.  
ENDLOOP.
```

- Fetches data from table ZCUSTOMER
- Displays data line by line in a simple list format.

Classical Reports are basic, easy to create, and ideal for simple data listing, but for advanced reporting, ALV or OO-based reports are preferred.

177. Difference between BAPI and RFC

Answer:

In SAP ABAP, BAPI and RFC are related but serve different purposes. Both enable remote communication, but their scope, usage, and standardization differ.

Key Points:

Feature	BAPI	RFC
Definition	Standardized Business Application Programming Interface for business objects.	Remote Function Call, a protocol to call function modules remotely.
Purpose	To perform business-level operations (create sales order, post invoice).	To execute any function module remotely.
Standardization	Predefined by SAP; follows Business Object Repository (BOR).	Not necessarily standardized; any remote-enabled FM can be called.
Usage	Mainly for integration and application-level operations.	Used for remote function execution, integration, or system communication.
Return Values	Returns structured messages in a RETURN table for errors/warnings.	May return simple values; error handling is manual.
Level	Business object level (high-level).	Function module level (low-level).
Remote-Enabled	Yes, always remote-enabled.	Must be remote-enabled explicitly.

Real-Time Example:

- BAPI: BAPI_SALESORDER_CREATEFROMDAT2 → Creates a sales order in SAP.
- RFC: Z_CALCULATE_DISCOUNT → Custom function called from another system.

"Calling BAPI

```
CALL FUNCTION 'BAPI_SALESORDER_CREATEFROMDAT2'
```

```
EXPORTING
```

```
  order_header_in = ls_order_header
```

```
TABLES
```

```
  return          = lt_return  order_items_in = lt_order_items.
```

"Calling RFC

```
CALL FUNCTION 'Z_CALCULATE_DISCOUNT' DESTINATION 'SAP_REMOTE'
```

185. Difference between LSMW and BDC

Answer:

In SAP ABAP, both LSMW and BDC are used for data migration, but they differ in approach, flexibility, and user-friendliness.

Key Points:

Feature	LSMW (Legacy System Migration Workbench)	BDC (Batch Data Communication)
Definition	A standard SAP tool for migrating data from legacy systems.	A technique/program to automate SAP transaction data entry.
Ease of Use	User-friendly; minimal coding required.	Requires ABAP programming or recording knowledge.
Methods Supported	Batch Input, Direct Input, BAPI, IDoc.	Session Method, Call Transaction Method.
Data Mapping	Provides GUI for field mapping and conversion rules.	Mapping is done manually in BDC program or using recording.
Error Handling	Built-in error logs; easier reprocessing.	Errors handled via SM35 (Session) or programmatically (Call Transaction).
Flexibility	Limited for complex logic; mostly standard migration.	Highly flexible; can implement custom validations and logic.
Use Case	Ideal for mass master data migration during SAP implementation.	Ideal for custom uploads, complex transactions, or legacy automation.

Real-Time Example:

- LSMW: Upload 1000 customer masters using mapping and batch input method.
- BDC: Upload custom financial documents using CALL TRANSACTION with field validations.

212. Difference between SAPScript and SmartForm

Answer:

SAPScript and SmartForm are both SAP tools used to design and print forms, but SmartForm is the modern, graphical alternative to SAPScript, offering more flexibility, less coding, and easier maintenance.

Key Points:

Feature	SAPScript	SmartForm
Layout	Text-based	Graphical (drag-and-drop)
Coding	Requires manual coding for logic	Minimal coding, uses auto-generated function modules
Ease of Use	Complex	User-friendly and intuitive
Tables & Loops	Manual handling needed	Supported natively with table nodes
Integration	Standard SAP modules	Standard SAP modules + flexible for custom forms
Upgrade-Friendly	Moderate	High, easier to maintain during upgrades
Graphics Support	Limited	Supports graphics, logos, and images easily
Printing Options	Printer, spool	Printer, spool, PDF, and email

Real-Time Example:

- SAPScript: Used in legacy systems to print sales orders with manual formatting of text and tables.
- SmartForm: Used to print customer invoices with company logo, dynamic tables for line items, and multi-language support.

237. Difference between SOAP and OData Service

Answer:

SOAP and OData are two types of services used for system integration in SAP, but they differ in protocols, data format, and usage scenarios.

SOAP is mainly for formal, enterprise-grade integrations, whereas OData (REST-based) is ideal for lightweight, mobile, and web applications.

Key Points:

Feature	SOAP Service	OData Service (REST)
Protocol	SOAP (Simple Object Access Protocol)	HTTP / REST
Data Format	XML only	JSON (mostly) or XML
State	Can be stateless or stateful	Stateless
Usage	Enterprise integrations, B2B systems	Fiori apps, mobile apps, lightweight web apps
Performance	Heavier, slower due to XML processing	Lightweight, faster with JSON
Standardization	Strict standards (WSDL, XSD)	Flexible, less strict
Error Handling	Uses SOAP Faults	Uses HTTP response codes
Tool in SAP	Configured via SOAMANAGER, consumed via proxy	Exposed via SAP Gateway / OData Services

Real-Time Example:

1. SOAP Service:
 - Expose BAPI_CUSTOMER_GETDETAIL2 as SOAP.
 - Used by external CRM system to fetch detailed customer info.
 - Response in XML, secure, reliable, suited for enterprise integration.
2. OData Service:
 - Expose ZCUSTOMER_SRV via SAP Gateway.
 - Fiori app consumes the service to display customer list in a lightweight way.
 - Response in JSON, fast and mobile-friendly.

16. New syntax

240. What is Inline Declaration?

Answer:

“Inline Declaration in ABAP is a modern syntax feature that allows you to declare and initialize a variable in a single statement.

It improves code readability, reduces clutter, and is widely used in S/4HANA ABAP programs.”

Key Points:

- Definition:
 - Inline declaration allows type inference by ABAP runtime.
 - No need to explicitly define the variable type using DATA; ABAP automatically determines the type from the assigned value.
- Syntax:

DATA(variable_name) = value.

or using LET in expressions:

DATA(result) = VALUE i(5 + 10).

- Benefits:
 1. Less boilerplate code → no separate type definition needed.
 2. Improved readability → variable appears exactly where it is used.
 3. Encourages functional programming style in ABAP.
 4. Works with all data types – elementary types, structures, and internal tables.

Real-Time Example:

1. Simple Inline Declaration:

```
DATA(lv_count) = 100. "lv_count automatically inferred as INT4
WRITE lv_count.
```

2. With Internal Table:

```
DATA(lt_customers) = VALUE TABLE OF zcustomers(
  ( kunnr = '1000' name = 'John' )
  WRITE: / ls_customer-kunnr, ls_customer-name.
Here, ls_customer is inline declared in the LOOP statement.
```

252. Differentiate between Classical ABAP and ABAP Objects.

Answer

Classical ABAP and ABAP Objects differ mainly in how programs are structured and how logic is reused.

1. Programming Approach

✓ Classical ABAP

Follows procedural programming — logic flows step-by-step in a sequence (reports, includes, forms).

✓ ABAP Objects

Follows object-oriented programming — logic is grouped into classes and objects with data + behavior.

2. Code Structure

✓ Classical ABAP

Program code is organized into:

- SUBROUTINES (FORM routines)
- Includes
- Function modules

✓ ABAP Objects

Program logic is encapsulated inside:

- Classes
- Attributes
- Methods
- Interfaces

3. Reusability

✓ Object-oriented ABAP supports inheritance, polymorphism, encapsulation, making reuse easy.

✓ Classical ABAP has limited reuse — mainly via INCLUDE or function calls.

4. Maintainability

- ✓ Classical ABAP → changes often mean modifying multiple reports.
- ✓ ABAP Objects → change one method/class and all consumers benefit.

5. Framework Support

Modern SAP frameworks (ALV OO, Web Dynpro, RAP, OData, BOPF, BRF+, classes for posting docs) are built on **ABAP Objects**, not classical ABAP.

Example:

CL_GUI_ALV_GRID, CL_SALV_TABLE, CL_ABAP_CONTEXT_INFO

6. Real-Time Example

Classical ABAP

A report posts FI documents using function modules.

If you want to reuse logic in OData service or batch job, you rewrite again.

ABAP Objects

Create a class ZCL_FI_POSTING with method POST.

Same method reused:

- ✓ Report
- ✓ WebDynpro
- ✓ OData API
- ✓ Background job

7. Future Adaptability

- SAP recommends OO programming for modern apps and Fiori/RAP development.
- Procedural is still used for simple reports or legacy maintenance.

18. CDS – Core Data Services

276. What is a CDS view and why is it used in S/4HANA?

A CDS view, or Core Data Services view, is an advanced way to define semantic views on database tables in SAP S/4HANA. It allows us to create reusable, database-optimized views with business logic, without writing complex ABAP code. CDS views push calculations to the HANA database, improving performance significantly."

Key Points:

1. CDS views are defined in **ABAP Development Tools (ADT)** using a DDL-style syntax.
2. They allow **annotations** for behavior, analytics, and UI consumption.
3. They can expose data directly to **Fiori apps, OData services, and analytical tools**.

Example:

- A CDS view ZCDS_SALES_ORDER can join tables like VBAK and VBAP, calculate totals, and be consumed in a Fiori sales app directly, reducing the need for additional ABAP reports.

"In short, CDS views in S/4HANA provide a modern, efficient, and reusable way to model business data on the HANA database for reporting, UI, and analytical purposes.

20. OData

311. What is OData and why is it used in SAP?

OData (Open Data Protocol) is a standard protocol used to expose and consume data over the web. In SAP, it allows Fiori apps, web applications, and external systems to access SAP data securely and efficiently.

Key Points:

1. Purpose in SAP
 - Acts as a bridge between SAP backend (ABAP, CDS, HANA) and frontend/UI applications.
 - Supports CRUDQ operations (Create, Read, Update, Delete, Query).
 - Standardized protocol, making integration with non-SAP systems easier.
2. How it works
 - Backend data (from tables, CDS views, or AMDP) is exposed as an OData service.
 - Frontend apps can consume this data via HTTP requests in JSON or XML format.
3. Real SAP Use Case:
 - Fiori apps like Manage Sales Orders or Invoice Processing consume OData services generated from CDS views.
 - External reporting tools can pull live SAP data using OData without direct database access.

In short, OData is the standard SAP protocol to expose backend data in a structured, secure, and frontend-consumable way for Fiori apps and integrations.